PRACTICAL NO. 5

Design and Implement Static Routing Using GNS3 software

Aim:

To design and implement static routing using two routers and two PCs in GNS3, and verify connectivity between the PCs.

Theory:

Static routing is a routing technique in which the network administrator manually adds routes to the router's routing table. The router uses these manually configured routes to forward packets to networks that are not directly connected.

The two LANs are:

- **Network 1:** 192.168.1.0
- **Network 2:** 192.168.2.0
- **Router-to-router network:** 10.0.12.0

Since R1 does not directly know about Network 2 and R2 does not directly know about Network 1, static routes are configured manually.

Static Route Syntax

ip route <destination-network> <subnet-mask> <next-hop-IP>

For example, on R1:

ip route 192.168.2.0 255.255.255.0 10.0.12.2

This means that R1 should forward packets destined for 192.168.2.0 to R2 at 10.0.12.2

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC1	Ethernet	192.168.1.2	255.255.255.0	192.168.1.1
R1	G0/0	192.168.1.1	255.255.255.0	—
R1	G0/1	10.0.12.1	255.255.255.252	—
R2	G0/0	10.0.12.2	255.255.255.252	—

R2	G0/1	192.168.2.1	255.255.255.0	—
PC2	Ethernet	192.168.2.2	255.255.255.0	192.168.2.1

Procedure

Step 1: Create the Network Topology in GNS3

1. Open **GNS3**.
2. Create a new project.
3. Add:
   1. 2 Cisco routers
   2. 2 VPCS
4. Rename them as **R1, R2, PC1, and PC2**.
5. Connect the devices as shown:

PC1 —— R1 —— R2 —— PC2

6. Start all devices.

Step 2: Configure Router R1

Open the console of R1 and enter:

```
enable
configure terminal

interface g0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

interface g0/1
ip address 10.0.12.1 255.255.255.252
no shutdown
exit
```

Configure Static Route on R1

R1 needs a route to the network connected to R2:

```
ip route 192.168.2.0 255.255.255.0 10.0.12.2
```

Then:

end

Step 3: Configure Router R2

Open the console of R2:

enable
configure terminal

interface g0/0
ip address 10.0.12.2 255.255.255.252
no shutdownexit

interface g0/1
ip address 192.168.2.1 255.255.255.0
no shutdownexit

Configure Static Route on R2

R2 needs a route to the network connected to R1:

ip route 192.168.1.0 255.255.255.0 10.0.12.1

Then:

end

Step 4: Configure PC1

Open the PC1 console.

Enter:

ip 192.168.1.2 192.168.1.1

Here:

- IP address = 192.168.1.2
- Subnet mask = 255.255.255.0
- Gateway = 192.168.1.1

Verify using:

show ip

Step 5: Configure PC2

Open the PC2 console:

ip 192.168.2.2 192.168.2.1

Step 6: Verify Router Interfaces

On R1 and R2, execute:

show ip interface brief

Step 7: Verify Static Routing

On R1:

show ip route

On R2:

show ip route

Step 8: Test Connectivity

First, test PC1 with its default gateway:

PC1> ping 192.168.1.1

Then test PC2:

PC1> ping 192.168.2.2

Conclusion:

Static routing was successfully configured between two routers and two PCs in GNS3. Connectivity between PC1 and PC2 was successfully verified using the ping command.